

TEAM 14

BMW GROUP'S FINANCIAL ANALYSIS

ACCOUNTING, FINANCE & CONTROL
PROF. MARIKA ARENA
A.Y. 2023/2024



**BMW
MOTORRAD**

**BMW
GROUP**
Financial Services

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INTRODUCTION

The BMW Group is a leading high-tech company recognized for its premium approach to mobility, offering high level products and services that aim at a luxury market.

The BMW Group's business spans three key segments:

- Automotive, featuring the premium brands BMW, MINI, and Rolls-Royce;
- Motorcycles, with BMW Motorrad positioned in the high-end market;
- Financial Services, credit financing and leasing automobiles and motorcycles (Alphabet).

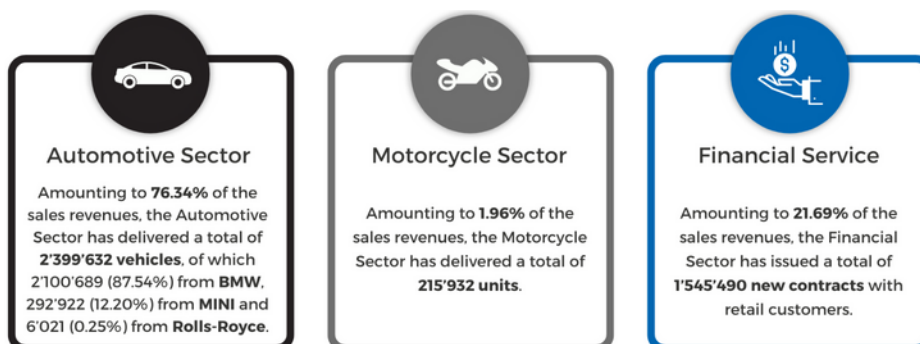


Image 1 - BMW Group's key segments

Value Chain

BMW Group's market trends are driven by consumer demand, with a focus on luxury. The company meets these demands with tailored, innovative vehicles including electric and hybrid models as well as traditional endothermic ones. This is feasible thanks to a structured value chain:

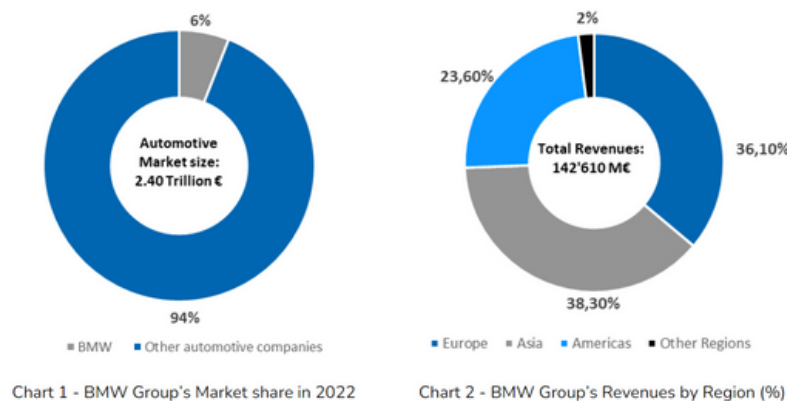
Support Activities	Infrastructure	Oliver Zipse, serving as CEO and Chairman of the Board of Management at BMW Group, oversees an infrastructure that features advanced manufacturing, comprehensive R&D, extensive dealership networks, and a dedication to sustainability in its global operations.				
	Human Resources	The BMW Group's team comprises individuals hailing from over 120 different countries, counting 149'475 employees in 2022. Occupying the 9th position in the Forbes World's best employers 2023, it attracts highly qualified personnel and aims to promote training programs, diversity and inclusivity.				
	Technology Dev.	The group has established 29 research and development centres distributed worldwide, counting 12 in Europe. R&D is key to success in the expanding EV market and exploring potential opportunities in hydrogen technologies.				
	Procurement	The Purchasing and Supplier Network accounts for global procurement and quality assurance of production materials, raw materials, capital assets and services, as well as managing the production of internal automobile parts.				
Primary Activities	Inbound logistics	Operations	Outbound logistics	Marketing & Sales	Products & Services	
	BMW Group's network involves 12'000'suppliers ca. located in 70 countries (while some are integrated by the company, the majority remain independent). Raw materials are predominantly sourced from Germany (33.7%), followed by China (17.8%) and Eastern Europe (17.7%). BMW Group is committed to sustainable practices and actively develops logistics goals aimed at climate-neutral transportation.	150'000 employees work in BMW Group within 31 production sites on 6 continents and R&D departments in 17 countries. To strengthen its global positioning, the company embraces a Local for Local strategy setting up manufacturing and assembly facilities in individual regions. BMW Group integrates Digital and Industry 4.0 innovations with advanced technologies and logistics operations management.	BMW Group has developed a worldwide robust sales and financial services network, further strengthened by their partnership with DHL through the "Connected Supply Chain" (CSC) system. The partnership between BMW and DHL is vital, as DHL handles deliveries for BMW in most of the markets they serve, while ensuring BMW a real-time oversight of its supply chain activities.	BMW's branding campaign has established a global reputation for quality and innovation, resulting in high customer loyalty. The company emphasises digital marketing and operates a global sales network with over 3'600 BMW, 1'600 MINI, and 150 Rolls-Royce dealerships. The Group is now reshaping its Sales & Marketing division to enhance collaboration with third-part-dealers and offer online purchasing with home delivery.	With strong brands such as BMW AG, MINI, Rolls-Royce and BMW Motorrad, the BMW Group excels in the premium car and motorcycle sectors. The competitive edge lies in exceptional post-sale services: expert maintenance, technical support, warranties, parts, and accessories. The company also provides credit financing and leasing options through their Financial Services segment in over 50 countries.	

Image 2 - BMW Group's Value Chain

¹ Datum sourced from the most recent BMW Group Sustainability Factbook available, published in 2019.

Leveraging its structure, BMW is strategically positioned in the ever-evolving automotive industry. The automotive industry is a highly competitive sector, and has important entry barriers, with key competitors such as Mercedes-Benz, Audi, Tesla and Porsche in the premium market. Rapid technological advancements and continuous shifting consumer preferences are driving innovation, with a growing market share on electric and autonomous vehicles. Peering into the future, hydrogen-powered vehicles are the subject of research and considered a sidekick to e-mobility. Additionally, regulatory changes regarding emissions standards and environmental sustainability are influencing the industry rules. Finally, industry wise internal strategies must had been reshaped in order to face different challenges, such as the COVID19, Ukraine conflict and Taiwan semiconductor crisis, which overall led to not negligible supply shortages that impacted on the overall supply chain structure as well as the deliveries' lead time and raw materials and energy costs.

Understanding and adapting to these external factors is crucial for BMW in order to maintain and increase its dominance in this premium automotive market segment.



Strategy

Once understood the external environmental challenges, it becomes important to align the internal levers and strategies in order to be flexible and promptly react.

BMW Group's strategy is a dynamic, ongoing process shaped by reviews by the Board of Management. This adaptability integrates corporate and product strategies into a cyclical planning system, fostering flexibility and responsiveness to market changes. The strategy, crafted by its leadership, fluidly combines the company's structure (*body*), values (*heart*), and adaptability (*mind*) to efficiently chase its ultimate ambitions.

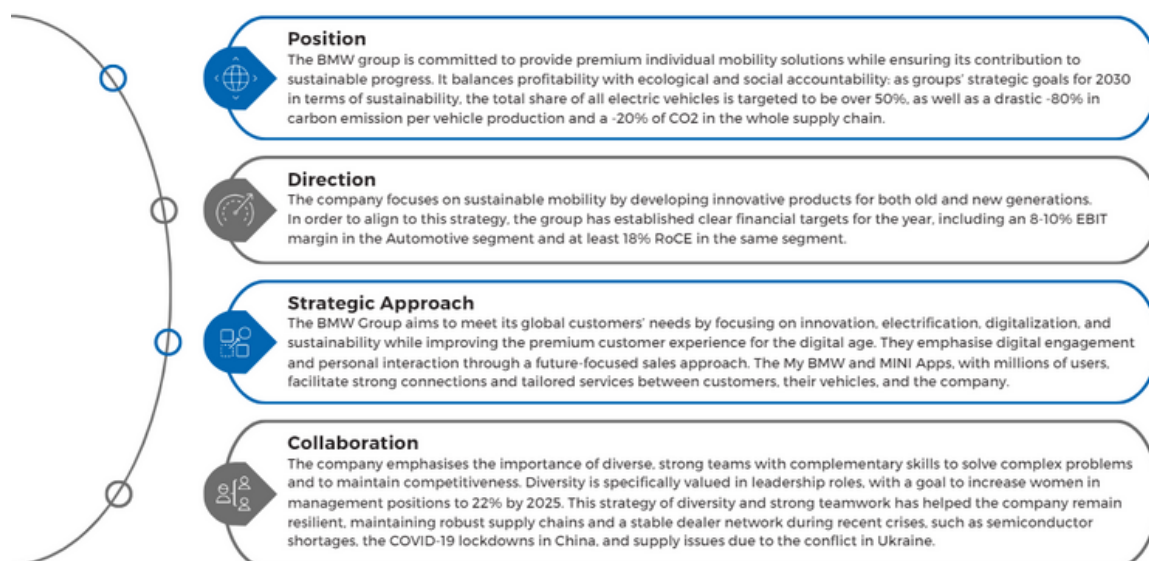


Image 3 - BMW Group's Strategy

LONGITUDINAL ANALYSIS

To understand the Group's performances, we conducted a financial analysis focusing on the profitability and liquidity indicators. The examination of the BMW Group's performances involved a thorough analysis of the annual reports, starting from 2019 until 2022 and evaluating trends over the 4 years.

Given the company's status as a publicly traded entity, the consolidated financial statements adhered to the International Financial Reporting Standards (IFRS). Data used for the internal analysis, including elements of horizontal analysis to show variations over time, are reported in [Appendix1], [Appendix2].

A set of both liquidity and profitability indicators were selected, aiming to cover all shareholders', stakeholders' and overall company perspectives:

Profitability	→ Return on Investment • Return on Assets • EBIT Margin • Asset Turnover Ratio → Return on Equity → Return on Capital Employed → Debt To Equity → Financial leverage - DuPont Approach
Liquidity	→ Quick Ratio and Current Ratio → Cash Flow to Debt Ratio → Capital Expenditure Coverage Ratio

Tab 1 - Selected indicators

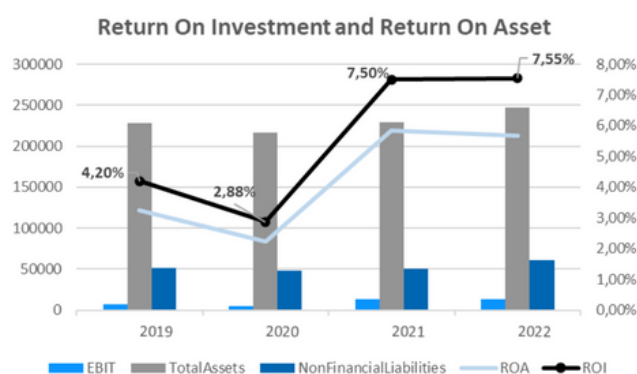
Profitability Indicators

RETURN ON INVESTMENT

$$ROI = \frac{EBIT}{Total Assets - Non Financial Liabilities}$$

The most significant indicator to measure the company's overall perspective is ROI, which monitors the ability to generate profit from the transformation of the input into output. As shown in the Graph 1, the indicator has grown significantly between 2019 and 2022, even if it had an important drop in 2020. By investigating the components of the ROI, the variation of the denominator (-4.95% between 2019 and 2020, +6.37% between 2020 and 2021, +3.86% between 2021 and 2022)² does not appear to be relevant in its overall change; on the other hand, the variations are in fact driven by the EBIT. In particular, between 2019 and 2020, EBIT has fallen by 34.83%: this is mainly caused by a reduction in revenues, not followed by a comparable reduction in cost of sales; the root cause can be identified in the COVID-19 Pandemic, which led to the closure of dealerships, resulting in a reduction of deliveries.

With a regaining in revenues (+12.37%, increasing from 98'990 to 111'239 M€) the group was able to recover the performances already in 2021; among the components affecting EBIT (+177.43%), another notable variation is to be found in Other Operating Income (+85.81%): as the company testifies in the 2021 Annual Report, it is "[...] primarily due to the income arising on the partial reversal of the provision relating to the antitrust proceedings of the EU Commission, which was recorded in the second quarter 2021".³



Graph 1 - Return on Investment and Return On Assets

² From now on, the data reported in the parenthesis will be presented in chronological order. Regarding variations over the years, they will be intended as (2020-2019, 2021-2020, 2022-2021).

³ The EU Commission accused BMW Group of conspiring to limit competition in the development of exhaust treatment systems for both petrol and diesel passenger cars. The amount of the provision exceeding the fine was reversed with the income statement effect in the year under report. The fine was paid in July 2021, thereby concluding the EU Commission's proceedings.

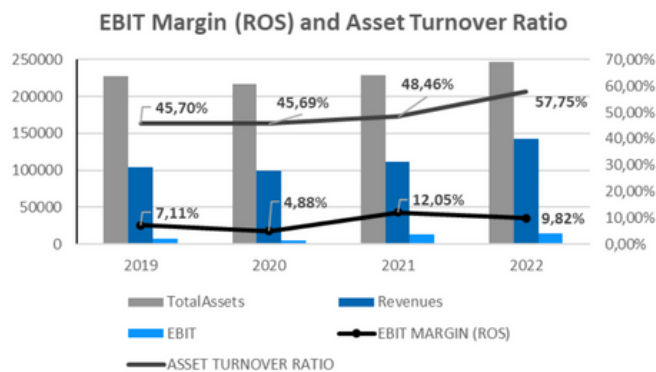
Having assessed a ROI level of 7.50% in 2021, the indicator stabilised by reaching 7.55% in 2022. Despite the growth of the indicator, the Group was unable to reach and outperform its WACC (12%).

Given that the NFL plays a negligible role in shaping the trend, ROI follows a similar trajectory to ROA, which can be dissected into two different components.

$$ROA = ROS \cdot Assets\ Turnover = \frac{EBIT}{Revenues} \cdot \frac{Revenues}{Total\ Assets}$$

In particular, ROS (EBIT Margin) is monitored as a strategic KPI by the BMW Group, therefore it has been taken into consideration for further analysis.

As can be observed in the formula, Revenues influence both ROS and ATR. After the previously mentioned contraction in sales of -5%, registered in 2020 compared to 2019, an important differential increase is shown in 2021 compared to 2020 (+12.37%) and an additional +28.20% in 2022.



Graph 2 - EBIT Margin and Asset Turnover Ratio

As described for ROI, also ROS is strongly driven by EBIT. For 2019 to 2021, the EBIT variation dominates the Revenue one. As a consequence, ROS follows EBIT's trend. In 2022 instead, compared to the previous years, Revenue variation dominates EBIT's, leading to the ROS' variation. This phenomenon can be explained by an increase in the Cost of Sales (+32.26%), which is related to limited availability of semiconductors, supply chain disruptions and higher prices for raw materials and energy, influenced by the Taiwan chips crisis and the Russia-Ukraine conflict.

BMW sets a target of 8-10% for the ROS of the Automotive Segment. By observing the group level only, the target does not always seem to be reached, and, by deepening the analysis at the Automotive Segment level (4.91%, 2.67%, 10.34%, 8.60%) [Appendix2], the same conclusion can be formulated: while in 2019 and 2020 the aimed range for the EBIT Margin is not achieved, in 2021 and 2022 the company did succeed. For the second component of the ROA breakdown formula, a parallel trend in Revenues and Total Assets can be observed. ATR is almost constant for the 2019/2021 period. For 2021-2022, instead, the ATR ratio faces a +9.29% increase driven by the Revenues, which have registered a stronger differential increase if compared to the Assets one. Despite ATR consistently displaying higher values than ROS, exerting a greater influence on the value of ROA, the latter's variations are primarily driven by ROS's fluctuations. The relative stability of NFL suggests that ROI variations, akin to ROA, are likely influenced by ROS.

RETURN ON EQUITY

$$ROE = \frac{Net\ Profit}{Equity}$$

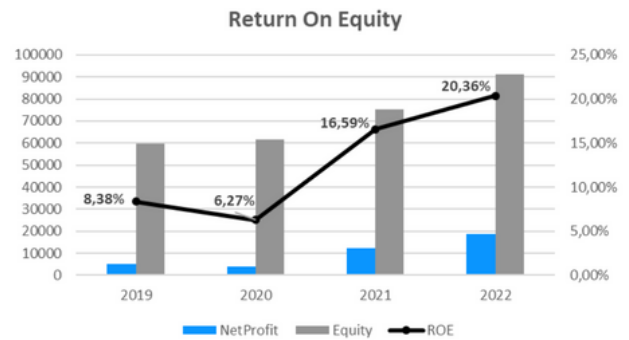
This indicator measures how much shareholders earned with their investment, and has shown a comprehensive increase from 2019-2022, except for 2019-2020 (8.38%, 6.27%, 16.59%, 20.36%). The variation of ROE, as represented in Graph 3, is mainly led by changes in Net Profit; overall, the latter follows the same trend as EBIT, with differences caused by the impact of Financial Results and Taxes. Between 2019 and 2020, whilst EBIT decreased by 26.64%, Net Profit declined by 23.20%: a positive Financial Result (from -293 to +392 M€) and a diminishing Effective Tax Rate⁴ slowed the decrease of the latter. In the following years, the increase of EBIT drove the increase of Net Profit, which was once again boosted by a combination of improved Financial Results (392, 2660 and 9510 M€) and lower Effective Tax Rates (26.14%, 22.40%, 20.96%). The trend of Equity, even if characterised by a sustained growth rate, is less accentuated when compared to Net Profit.

⁴ Higher in 2019 as a result of the non-deductible items for tax purposes related to the provision for the EU antitrust proceeding already mentioned in the ROI analysis.

Nevertheless, it is interesting to investigate the causes: from 2020 to 2021 an increase of 22.13% can be observed, and it is mainly due to an increase of Revenue reserves of over 12B€ (+20.41%).

Instead, the significant increase of 21.50% between 2021 and 2022 is related to a crucial event for the BMW Group, affecting its structure: on 11 February 2022, the BMW Group increased its shareholding in the BMW Brilliance joint venture from 50% to 75% with the acquisition of a further 25% of shares. This acquisition was motivated by the strategic purpose of strengthening the group's long-term collaboration with a Chinese partner and expanding production capacities in the area, in line with the "local for local" principle. BMW Brilliance has therefore been fully consolidated as a subsidiary. This event caused a significant increase in Minority (non-controlling) Interest.⁵ Other than that, the acquisition of Brilliance impacted further voices, including Other Financial Results, which drove Financial Results in 2022.

The joint effect of the changes in the numerator and denominator resulted in a positive trend of the indicator, granting promising expectations for shareholders.



Graph 3 - Return on Equity

RETURN ON CAPITAL EMPLOYED

$$RoCE = \frac{EBIT}{\text{average amount of capital employed}} = \frac{EBIT}{(\text{equity} + \text{long term financial debt})}$$

Among the overall company perspective's indicators, RoCE has been included in the analysis because it is considered a Key Performance Indicator by BMW Group, providing crucial insights on the profitability of capital employed and business operations.



Graph 4 - Return on Capital Employed

Over the period of analysis, the average amount of capital employed has been relatively stable (-1.26%, +6.64%, +5.30%). Once again, the indicator is driven by the EBIT, resulting in the trend reported in Graph 4. BMW Group aims for at least 18% RoCE in the automotive sector and evaluates this segment's data using a different Capital Employed definition, "calculated as the sum of intangible assets, property, plant and equipment and net working capital, the latter comprising inventories as well as trade

receivables less trade payables". Looking at the results reported in the 2022 Annual Report, the indicator has been 29% in 2019, 12.7% in 2020, 24% in 2021 and 18.1% in 2022. Therefore, except for the year 2020 (strongly impacted by the Covid-19 Pandemic), the BMW Group consistently achieved the proposed goal.

DEBT TO EQUITY

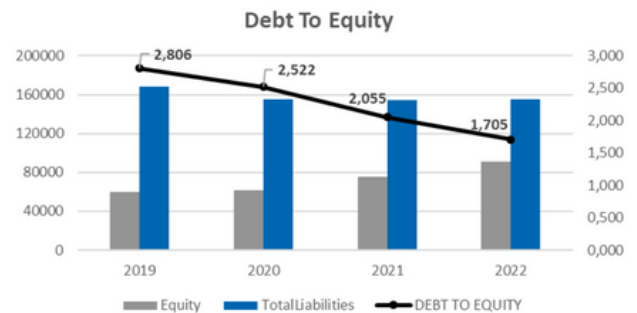
$$\text{Debt to Equity} = \frac{\text{Total Liabilities}}{\text{Equity}} = \frac{\text{Current Liabilities} + \text{Non Current Liabilities}}{\text{Equity}}$$

The increasing trend of Equity described in the previous paragraphs also affects another relevant indicator that gives an insight into the structure of the company's financing: Debt to Equity. Total Liabilities, the other component of the indicator, after being subjected to a 7.73% decrease from 2019 to 2020,⁶ stabilised to a constant range. As a consequence, over the years BMW Group has become less and less exposed to debt.

⁵ As stated in the annual report: "The 25% noncontrolling interests of other shareholders in the subsidiary BMW Brilliance represent a significant item in the BMW Group's balance sheet".

⁶ It is to say that in 2019 there has been a peak in Financial Liabilities caused by new bonds issued in the US and China.

A declining Debt-to-Equity ratio over time indicates financial stability. On the one hand, this shows that the company is effectively managing its debt levels and maintaining a sustainable capital structure. On the other hand, it may indicate that the Group is not fully utilising its financial leverage. Benchmarking the value with the primary competitors will help determine which is the case.



Graph 5 - Debt To Equity

FINANCIAL LEVERAGE - DUPONT APPROACH

After studying Third Part Liabilities over Equity, it is feasible to further deepen the analysis by monitoring the other Financial Leverage components. For that scope, the DuPont approach is adopted:

$$ROE = NPM \cdot ATR \cdot \left(1 + \frac{TPL}{Equity}\right)$$

Looking at the formula, all the voices have already been computed and studied, except for the Net Profit Margin. The latter had a significant increase over the years: starting from 4.82% in 2019, it slowed down in 2020 (3.90%) and rose again to 11.20% in 2021 and 13.03% in 2022. Considering that TPL/Equity decreased over the years, improved performances in NPM and ATR were the drivers of ROE's enhancement.

Liquidity Indicators

QUICK RATIO AND CURRENT RATIO

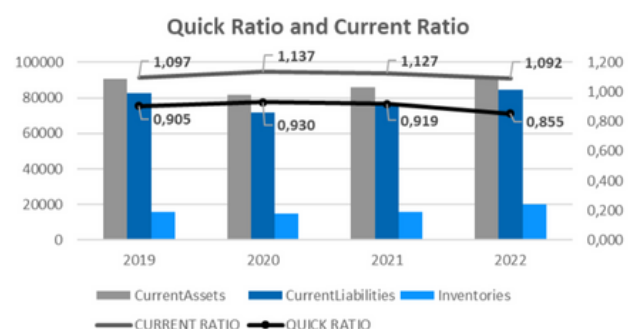
To monitor Liquidity performances, Quick Ratio and Current Ratio were the first indicators to be analysed. In particular, to be more conservative and adhering to the *Prudence Accounting Principle*, the Quick Ratio was highlighted. Doing that, a less severe approach was followed, shaping the numerator to consider Assets readily available to pay off Current Liabilities as *Current Assets - Inventories*, instead of *Cash + Short Term Investment + Receivables*, as for the more severe approach.

$$Quick\ Ratio = \frac{Current\ Assets - Inventories}{Current\ Liabilities}$$

$$Current\ Ratio = \frac{Current\ Assets}{Current\ Liabilities}$$

As seen in Graph 6, the Current Ratio has been higher than 1 over the period of study, proving a strong ability of the Group to pay its Current Liabilities off its Current Resources. However, the Quick Ratio values fluctuate around 0.9, showing that Inventories play a significant role in that.

The most significant gap between the two indicators can be observed in 2022: during that year, in fact, Inventories showed a relevant increase in Finished Goods and Raw Materials; additionally, 2022 registered the lowest level for both indicators driven by an increase of Trade Payables (which include for the first time 2'782M€ related to BMW Brilliance) and Other Liabilities. Even though the Quick Ratio did not reach the desirable value of 1, no liquidity issues were reported over the period of study. Moreover, as a further reduction of risk, the Group secured an 8B€ syndicated credit line from a consortium of international banks.



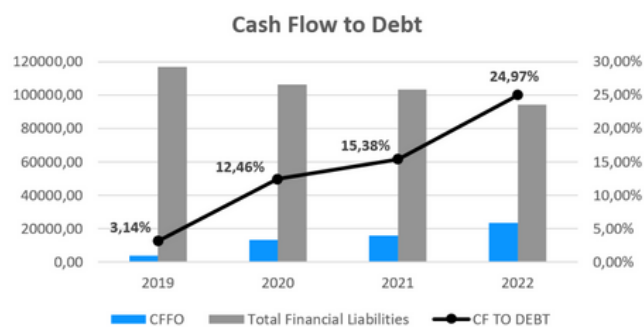
Graph 6 - Quick and Current Ratio

CASH FLOW TO DEBT

$$CF\ to\ Debt = \frac{Cash\ Flows\ From\ Operating\ Activities}{Current\ Financial\ Liabilities + Non\ Current\ Financial\ Liabilities}$$

Having considered the ability of the Group to cover its Current Liabilities, the following indicator aims at measuring its ability to cover its entire Financial Debt, using the Cash Flows coming from Operating Activities. Total Financial Liabilities between 2020 and 2019 follow a similar trend as the one previously seen for Total Liabilities, registering a decrease of -8.88%.

During the following years, on the other hand, while Total Liabilities were almost constant, Total Financial Liabilities continued decreasing (-2.74%, -8.96%). Even if this continuous decrease contributes to the indicator's growth, the latter is mainly driven by the numerator. A massive increase in CFFO (+261.85%) between 2019-2020 is mainly related to a variation in Change in Receivables from Sales Financing which goes from -3'560M€ to 4'192M€, caused by the lower vehicle inventories at dealerships due to the pandemic-related drop. In the subsequent years, the growth continued (+20.10%, +47.81%), led by increasing Profit before Tax. The continuous growth of Cash Flow to Debt grants the Group a stronger Financial Position, and long-term Financial Wealth. However, as done in the benchmarking analysis, it is of utter importance to compare this indicator with the ones of other industry competitors.



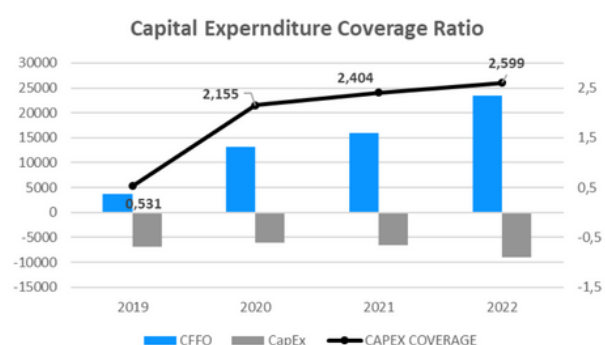
Graph 7 - Cash Flow to Debt

CAPITAL EXPENDITURE COVERAGE

The increase observed above in CFFO also affects the CapEx⁷ Coverage Ratio. This indicator, which measures the Group's ability to fund its Capital Expenditures from Internal Cash Flows, has been included considering the importance given to investing activities by BMW.

$$CapEx\ Coverage\ Ratio = \frac{Cash\ Flows\ From\ Operating\ Activities}{CapEx}$$

Graph 8 displays promising values from 2020, standing above 2. 2019 was the only year in which the Group was not able to self-finance, indicating that the company needed to use external sources, such as debt, to fund a portion of its CapEx, which can increase financial risk. In 2020 both the numerator and the denominator contribute to the increase of the ratio, with slowing down investments due to the Covid-19 Pandemic and growing CFFO. Instead, in 2021 and 2022 the trend is driven by CFFO, which are able to cover the increasing CapEx.



Graph 8 - CapEx Coverage Ratio

⁷ CapEx has been proxied by Total investment in intangible assets and property, plant and equipment.

BENCHMARKING ANALYSIS

To gain a comprehensive understanding of BMW's performance and its market strategy, it is essential to benchmark its financial results against comparable competitors. Since BMW's core business is represented by the automotive sector, the following Groups are taken into consideration:

Condensed 2022 data for comparison evaluation						
COMPANY	BMW Group	Mercedes-Benz Group	Audi Group	Volkswagen Group	Toyota Group	Tesla Group
Closing period	December, 31	December, 31	December, 31	December, 31	March, 31	December, 31
Currency	M€	M€	M€	M€	BillionYen*	M\$**
Revenues	142'610	150'017	61'753	279'232	37'154.2 (285'344 M€)	81'462 (77'470 M€)
EBIT	13'999	20'458	7'550	22'124	2'725 (20'928 M€)	13'656 (12'986 M€)
Net Profit	18'582	14'809	7'116	15'836	2'451.3 (18'826 M€)	12'587 (11'970 M€)
Number of Employees	149'475	171'382	87'342	669'275	375'235	127'855
Deliveries (Automotive segment)	2'399'632	2'040'700	1'638'638	7'957'288	8'822'000	1'313'851
Target Market	Luxury cars	Luxury cars	Luxury cars	Luxury cars; Middle-Low segment cars	Middle segment cars	Luxury-electric-only cars
Business Units	Automotive; Motorcycle; Financial	Automotive; Financial	Automotive; Motorcycle; Financial	Automotive; Commercial cars; Financial	Automotive; Services	Automotive; Energy service
Reporting Standard	IFRS	IFRS	IFRS	IFRS	IFRS	US GAAP

[*Average EURJPY 2022: 1 yen=0,00768 eur; **Average EUR/USD 2022: 1 dollar = 0.951 eur]

Tab 2 - Benchmarking Comparison Table

Mercedes and BMW are closely comparable in all of the categories shown in the table. Exactly as BMW, Mercedes serves the European High/Luxury automotive segment, with a similar market share in terms of revenues and deliveries. The comparison has grown more intriguing in 2021 and 2022, following the former Daimler's decision to separate the automotive business from truck and commercial vehicles.⁸

Volkswagen, as Europe's leading car manufacturer, is an important reference point, and the Audi Group, which has been a subsidiary of Volkswagen since 1964, should also be observed closely. Compared to BMW, Volkswagen's revenues are nearly double, it has four times as many employees and car deliveries, and serves a lower market segment.

AUDI is smaller in size compared to BMW but competes in the same luxury segment. BMW's brand portfolio includes Rolls Royce, Mini, and a motorbike division, whereas AUDI oversees Bentley (owned by AUDI starting from 2022, previously under Volkswagen), Lamborghini, and Ducati. Therefore, AUDI has been selected for financial benchmarking instead of Volkswagen.

Toyota was considered a benchmark for the sector but was ultimately excluded due to geographical differences, scale discrepancy, different market segments and a mismatch in financial year-end dates.

Tesla was excluded from a comparison with BMW because it offers fewer car models, only makes electric vehicles, has a smaller presence in Europe, is much younger as a company, and follows different accounting standards (US GAAP vs IFRS).

Data used in the benchmarking analysis is documented in [Appendix3]. With the exclusion of ROCE (more pertinent to BMW's internal strategy), the indicators utilised previously have been compared among the chosen Groups.

⁸ With effect in the commercial register from 9 December 2021 the former Daimler AG Group was separated into the remaining Mercedes-Benz Group for passenger vehicles and the new Daimler Truck Group for commercial vehicles, each with their strategy.

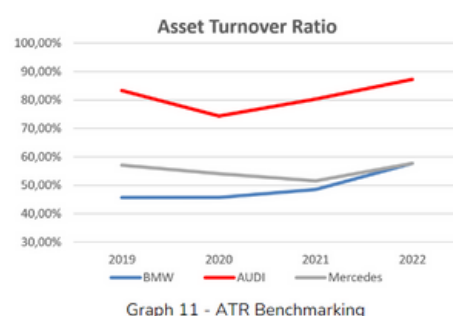
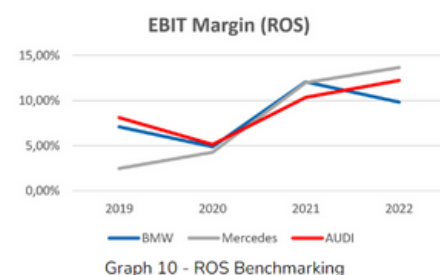
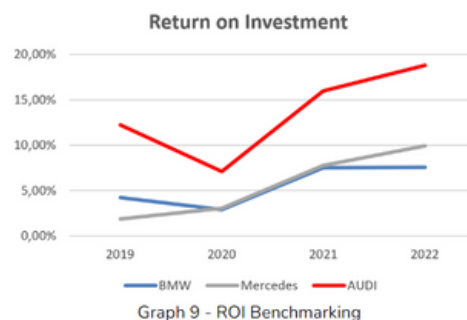
RETURN ON INVESTMENT

In order to get an overall insight of the chosen companies, it is suitable to start the benchmarking analysis with the study of ROI.

The values of BMW and Mercedes are quite similar over the years while the Audi Group shows outstanding performances, being the most profitable company. Except for the differences in figures, BMW and Audi follow a similar path. Considering that the Total Assets remained quite constant for each company over the years, the differences should be found in the ROI numerator. The main event affecting EBIT in 2020 has been the COVID-19 Pandemic: all the three companies resulted in reduced deliveries and revenues that were translated into a decreased EBIT for BMW and Audi. Mercedes was able to overcome these effects because of an improved sales structure with reduced costs and reduced Other Operating Expenses,⁹ resulting in improved performances also between 2019 and 2020. In the following years the three Groups bettered ROI values with BMW being slightly surpassed.

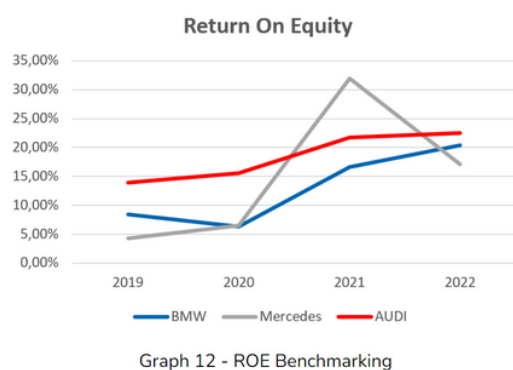
For sake of consistency with the internal analysis, also ROS and ATR were compared among the three. The latter, as seen for BMW, has remained quite constant also for the others.

ROS values are comparable among the Groups: between 2019 and 2020 Mercedes is the only one to improve the ratio for the previously mentioned reasons; subsequently, after resembling trends between 2020 and 2021, BMW is the one who follows a different direction: due to a stronger increase in Cost of Sales than in Revenues in 2022, the indicator has decreased.



RETURN ON EQUITY

As seen, in 2020 all the three groups were impacted by the Pandemic, resulting in a decrease of Revenues: this is translated in negative ROE trend for BMW, while Audi (with a reduction of Total Equity driven by Retained Earnings) and Mercedes (which managed to double Net Profit despite Revenues' shrinkage) experienced a slight increase in the ratio.



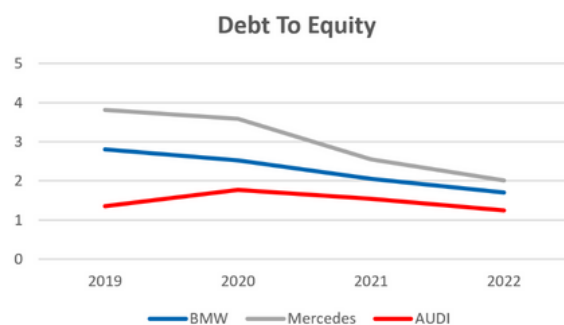
Over the following years, BMW Group and Audi showed a similar growth trend; Mercedes was subjected to a peak in 2021 due to a significant increase in Net Profit boosted by the recovery from the pandemic and a positive impact of the Group's restructuring with the previously mentioned spin-off of the Daimler commercial vehicle business. Following the peak, Mercedes' growth interrupted, bringing the three companies to a comparable level in 2022. With an exception for 2021, Audi overperformed the competitors, showing more promising performances for shareholders.

⁹ Which registered a high value the prior year, in connection with ongoing proceedings relating to Mercedes-Benz diesel vehicles in various regions and markets.

DEBT TO EQUITY

As seen in the internal analysis, BMW had a declining Debt to Equity ratio over time. Graph 13 shows that the BMW trend is in line with the other two companies.

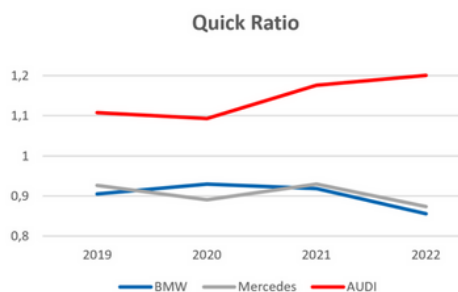
Between 2019 and 2020, Audi was the only one seeing an increment in the indicator with the combination of an increasing Debt and diminishing Equity: the reason is to be found in the agreement with Volkswagen that led to an impact on Audi's Retained Earnings, and as a consequence required funding through new Financial Liabilities. In spite of that, the Group was the least exposed over the period of analysis, with BMW consistently positioned in the middle between the other two.



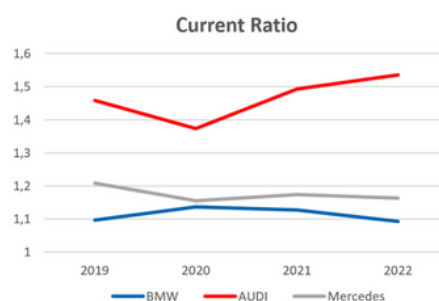
Graph 13 - Debt to Equity Benchmarking

Liquidity Indicators

QUICK RATIO AND CURRENT RATIO



Graph 14 - Quick Ratio Benchmarking



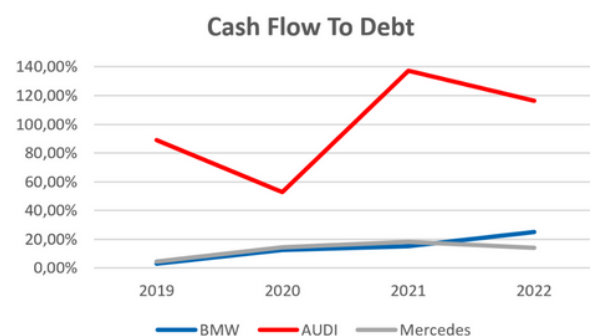
Graph 15 - Current Ratio Benchmarking

To benchmark the liquidity position of the Groups, Quick Ratio and Current Ratio have been analysed. Both the indicators showed an analogous trend, meaning that the Inventories did not significantly impact the pattern. As it is possible to infer, Audi proved a stronger ability in paying its current liabilities with its most liquid assets. While Mercedes always outperformed BMW in Current Ratio, the Quick Ratio trends of the two are perfectly aligned, floating around a satisfactory value of 0.9. Even though slightly below the desirable value of 1, similar to what was seen previously for BMW, Mercedes also maintained Extra Liquidity Reserves through facilities established with various banks.

CASH FLOW TO DEBT

As for the ability to cover Current Liabilities, Audi Group achieved higher performances also in covering its entire financial debt. Cash Flow From Operating Activities was almost equal or even higher than the Total Financial Liabilities (TFL) over all the years except 2020, when the latter increased and CFFO had a reduction "*principally due to lower earnings as a result of the Pandemic*".

BMW and Mercedes indicators progress with analogous values, but while the former Group is continuously improving its positioning, as seen in the previous analysis, the latter faced a reduction between 2021 and 2022: despite TFL continued their declining trend, the numerator had a more significant reduction, led by the decrease in Profit from continuing operations before taxes (in 2021, in fact, this voice included also discontinued operations related to Daimler Trucks & Buses).

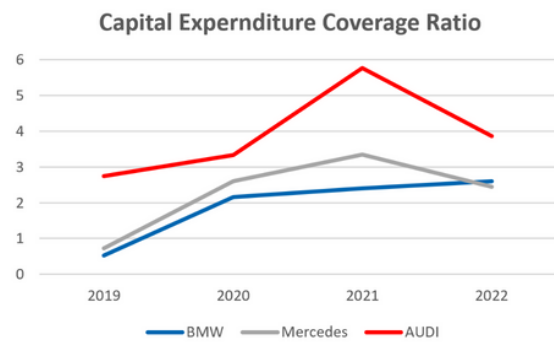


Graph 16 - Cash Flow to Debt Benchmarking

CAPITAL EXPENDITURE COVERAGE

Capital Expenditure coverage improved for all the three companies between 2019 and 2021.

Between 2021 and 2022 both Audi and Mercedes decreased their performances after reaching the peak in 2021, while BMW kept following a constant slower growth and eventually reached Mercedes. Over the years BMW has shown a weaker ability to fund its CapEx from Internal Cash Flows, but demonstrated a constant improvement.



Graph 17 - CapEx Coverage Benchmarking

CONCLUSION

Despite economic challenges and disruptions in the supply chain due to geopolitical factors and semiconductor shortages, the BMW Group proved to be a resilient and strategically positioned company in the competitive premium automotive sector. After the impact of the COVID-19 pandemic in 2020, the company rebounded strongly, showcasing positive trends across various financial indicators. The internal analysis of the Group highlights increasing performances on all the dimensions of analysis, with the exception for Current Ratio and Quick Ratio. These two indicators showed a slightly decreasing path over the last two years, anyway maintaining satisfactory values. Moreover, examining the target values established for EBIT Margin and RoCE the Group showcased its capacity to achieve strategic objectives.

The benchmarking analysis against key competitors Mercedes and Audi revealed that BMW maintained a competitive standing. The latter's performance closely aligned with Mercedes, both demonstrating adaptability to challenging market conditions. Audi, however, outperformed its competitors on ROI, Debt to Equity, and the analysed liquidity indicators: it is noteworthy that, as a subsidiary of the prominent Volkswagen Group, it potentially benefited from more advantageous ways of financing.

BMW's strategic responses to external challenges have proven effective, with positive results for its stakeholders and shareholders. The consistent improvement in key financial indicators, such as ROI and ROE, indicates a well-managed and resilient business model. Furthermore, BMW's commitment to financial prudence - demonstrated through declining debt levels and stable liquidity positions - instils confidence in the company's long-term stability. The results for the first three quarters of 2023 appear to confirm the growth trend: notably, an uptrend in the electric-vehicles market has been registered, with a nearly doubled nine-month delivery volume (246867 vs 128195, +92.6%).

Looking ahead, BMW's future prospects appear promising, guided by a customer-centric approach and a commitment to innovation and sustainability. Following its "North Star" - *We move body, heart and mind* - the Group positions itself at the forefront of change, envisioning a future that is electric, digital, and circular. BMW's forward looking approach, strategic initiatives and resilience position the company as an appealing investment opportunity.

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APPENDIX

Appendix 1

BMW GROUP - Balance Sheet

BALANCE SHEET					HORIZONTAL ANALYSIS		
[MLN €]	2022	2021	2020	2019	2022-2021	2021-2020	2020-2019
ASSETS							
Intangible assets	21776	12980	12342	11729	67,77%	5,17%	5,23%
Property, plant and equipment	32126	22390	21850	23245	43,48%	2,47%	-6,00%
Leased products	42820	44700	41995	42609	-4,21%	6,44%	-1,44%
Investments accounted for using the equity method	420	5112	3585	3199	-91,78%	42,59%	12,07%
Other investments	1351	1241	735	703	8,86%	68,84%	4,55%
Receivables from sales financing	50368	51712	48025	51030	-2,60%	7,68%	-5,89%
Financial assets	3073	1715	2644	1370	79,18%	-35,14%	92,99%
Deferred tax	1758	2202	2459	2194	-20,16%	-10,45%	12,08%
Other assets	1030	1302	1216	1325	-20,89%	7,07%	-8,23%
Non-current assets	154722	143354	134851	137404	7,93%	6,31%	-1,86%
Inventories	20005	15928	14896	15891	25,60%	6,93%	-6,26%
Trade receivables	4127	2261	2298	2518	82,53%	-1,61%	-8,74%
Receivables from sales financing	35340	35705	36252	41407	-1,02%	-1,51%	-12,45%
Financial assets	5164	5800	5108	5955	-10,97%	13,55%	-14,22%
Current tax	1096	1529	606	1209	-28,32%	152,31%	-49,88%
Other assets	9602	8941	9110	11614	7,39%	-1,86%	-21,56%
Cash and cash equivalents	16870	16009	13537	12036	5,38%	18,26%	12,47%
Current assets	92204	86173	81807	90630	7,00%	5,34%	-9,74%
Total assets	246926	229527	216658	228034	7,58%	5,94%	-4,99%
EQUITY AND LIABILITIES							
Subscribed capital	663	661	660	659	0,30%	0,15%	0,15%
Capital reserves	2432	2325	2199	2161	4,60%	5,73%	1,76%
Revenue reserves	85425	71705	59550	57667	19,13%	20,41%	3,27%
Accumulated other equity	-117	-325	-1518	-1163	-64,00%	-78,59%	30,52%
Treasury shares	-1278	-	-	-	-	-	-
Equity attributable to shareholders of BMW AG	87125	74366	60891	59324	17,16%	22,13%	2,64%
Minority (non-controlling) interest	4163	766	629	583	443,47%	21,78%	7,89%
Equity	91288	75132	61520	59907	21,50%	22,13%	2,69%
Pension provisions	339	1247	3693	3335	-72,81%	-66,23%	10,73%
Other provisions	8445	7206	6488	5788	17,19%	11,07%	12,09%
Deferred tax	2765	1458	509	632	89,64%	186,44%	-19,46%
Financial liabilities	53469	62342	67390	70647	-14,23%	-7,49%	-4,61%
Other liabilities	6199	5676	5095	5100	9,21%	11,40%	-0,10%
Non-current provisions and liabilities	71217	77929	83175	85502	-8,61%	-6,31%	-2,72%
Other provisions	7316	6748	7494	7421	8,42%	-9,95%	0,98%
Current tax	1224	921	747	963	32,90%	23,29%	-22,43%
Financial liabilities	40727	41121	38986	46093	-0,96%	5,48%	-15,42%
Trade payables	14120	10932	8644	10182	29,16%	26,47%	-15,11%
Other liabilities	21034	16744	16092	17966	25,62%	4,05%	-10,43%
Current provisions and liabilities	84421	76466	71963	82625	10,40%	6,26%	-12,90%
Total equity and liabilities	246926	229527	216658	228034	7,58%	5,94%	-4,99%

BMW GROUP - Income Statement

INCOME STATEMENT					HORIZONTAL ANALYSIS		
[MLN €]	2022	2021	2020	2019	2022-2021	2021-2020	2020-2019
Revenues	142610	111239	98990	104210	28,20%	12,37%	-5,01%
Cost of sales	-118042	-89253	-85408	-86147	32,26%	4,50%	-0,86%
Gross profit	24568	21986	13582	18063	11,74%	61,88%	-24,81%
Selling and administrative expenses	-10616	-9233	-8795	-9367	14,98%	4,98%	-6,11%
Other operating income	1377	1702	916	1031	-19,10%	85,81%	-11,15%
Other operating expenses	-1330	-1055	-873	-2316	26,07%	20,85%	-62,31%
Profit / loss before financial result	13999	13400	4830	7411	4,47%	177,43%	-34,83%
Result from equity accounted investments	-100	1520	920	136	-106,58%	65,22%	576,47%
Interest and similar income	422	135	116	179	212,59%	16,38%	-35,20%
Interest and similar expenses	251	-165	-458	-499	-252,12%	-63,97%	-8,22%
Other financial result	8937	1170	-186	-109	663,85%	729,03%	-70,64%
Financial result	9510	2660	392	-293	257,52%	578,57%	233,79%
Profit / loss before tax	23509	16060	5222	7118	46,38%	207,55%	-26,64%
Income taxes	-4927	-3597	-1365	-2140	36,98%	163,52%	-36,21%
Profit / loss from continuing operations	-	-	3857	4978	-	-	-
Profit / loss from discontinued operations	-	-	-	44	-	-	-
Net profit / loss	18582	12463	3857	5022	49,10%	223,13%	-23,20%
Attributable to minority (non-controlling) interest	641	81	82	107	691,36%	-1,22%	-23,36%
Attributable to shareholders of the BMW AG	17941	12382	3775	4915	44,90%	228,00%	-23,19%
Basic earnings per share of common stock in €	27,31	18,77	5,73	7,47	45,50%	227,57%	-23,29%
Basic earnings per share of preferred stock in €	27,33	18,79	5,75	7,49	45,45%	226,78%	-23,23%
Dilutive effects	-	-	-	-	-	-	-
Diluted earnings per share of common stock in €	27,31	18,77	5,73	7,47	45,50%	227,57%	-23,29%
Diluted earnings per share of preferred stock in €	27,33	18,79	5,75	7,49	45,45%	226,78%	-23,23%

BMW GROUP - Cash Flow Statementt

CASH FLOW STATEMENT					HORIZONTAL ANALYSIS		
[MLN €]	2022	2021	2020	2019	2022-2021	2021-2020	2020-2019
Profit / loss before tax	23509	16060	5222	7118	46,38%	207,55%	-26,64%
Income taxes paid	-4512	-3217	-1605	-3389	40,25%	100,44%	-52,64%
Interest received	453	130	163	91	248,46%	-20,25%	79,12%
Other interest and similar income / expenses	-206	78	104	51	-364,10%	-25,00%	103,92%
Depreciation and amortisation of tangible and intangible assets	8566	6495	6139	6017	31,89%	5,80%	2,03%
Other non-cash income and expense items	-7392	-713	99	-200	-936,75%	-820,20%	149,50%
Result from equity accounted investments	100	-1520	-920	-136	106,58%	-65,22%	-576,47%
Gain / loss on disposal of tangible and intangible assets and marketable securities	-	-	-	4	-	-	-
Change in leased products	2794	-1282	-1016	-3825	317,94%	-26,18%	73,44%
Change in receivables from sales financing	1888	965	4192	-3560	95,65%	-76,98%	217,75%
Changes in working capital	-523	1355	-996	-1117	-138,60%	236,04%	10,83%
Change in inventories	-115	-563	370	-1560	79,57%	-252,16%	123,72%
Change in trade receivables	-939	119	160	14	-889,08%	-25,63%	1042,86%
Change in trade payables	531	1799	-1526	429	-70,48%	217,89%	-455,71%
Change in provisions	816	-1256	1115	1512	-164,97%	-212,65%	-26,26%
Change in other operating assets and liabilities	-1970	-1181	754	1096	-66,81%	-256,63%	-31,20%
Cash inflow / outflow from operating activities	23523	15914	13251	3662	47,81%	20,10%	261,85%
Total investment in intangible assets and property, plant and equipment	-9050	-6619	-6150	-6902	36,73%	7,63%	-10,90%
Proceeds from subsidies for intangible assets and property, plant and equipment	102	43	53	50	137,21%	-18,87%	6,00%
Proceeds from the disposal of intangible assets and property, plant and equipment	103	40	34	32	157,50%	17,65%	6,25%
Expenditure for acquisitions, net of cash acquired	3587	-	-	-	-	-	-
Expenditure for investment assets	-117	-158	-176	-1598	-25,95%	-10,23%	-88,99%
Proceeds from the disposal of investment assets and other business units	32	263	1328	1087	-87,83%	-80,20%	22,17%
Investments in marketable securities and investment funds	-649	-271	-925	-775	139,48%	-70,70%	19,35%
Proceeds from the sale of marketable securities and investment funds	1220	302	2200	822	303,97%	-86,27%	167,64%
Cash inflow / outflow from investing activities	-4772	-6400	-3636	-7284	-25,44%	76,02%	-50,08%
Payments into equity	85	103	28	33	-17,48%	267,86%	-15,15%
Treasury shares acquired	-1278	-	-	-	-	-	-
Payment of dividends to shareholders of BMW AG	-3827	-1253	-	-	205,43%	-	-
Payment of dividends to non-controlling interests	-1455	-24	-	-	5962,50%	-	-
Payment of dividends for the previous year	0	-	-1671	-2366	-	-	-
Intragroup financing and equity transactions	-	-	-	-	-	-	-
Interest paid	-222	-217	-275	-199	2,30%	-21,09%	38,19%
Proceeds from issue of non-current financial liabilities	16050	18811	164478	150517	-14,68%	-88,56%	9,28%
Repayment of non-current financial liabilities	-26102	-26434	-171532	-143500	-1,26%	-84,59%	19,53%
Change in other financial liabilities	-1235	2279	718	305	-154,19%	217,41%	135,41%
Cash inflow / outflow from financing activities	-17984	-6735	-8254	4790	-167,02%	18,40%	-272,32%
Effect of exchange rate on cash and cash equivalents	104	-307	180	-28	133,88%	-270,56%	742,86%
Effect of changes in composition of Group on cash and cash equivalents	-10	-	-40	-83	-	-	-
Change in cash and cash equivalents	861	2472	1501	1057	-65,17%	64,69%	42,01%
Cash and cash equivalents as at 1 January	16009	13537	12036	10979	18,26%	12,47%	9,63%
Cash and cash equivalents as at 31 December	16870	16009	13537	12036	5,38%	18,26%	12,47%

BMW AUTOMOTIVE SEGMENT - Balance Sheet

BALANCE SHEET (Automotive Segment)					HORIZONTAL ANALYSIS		
[MLN €]	2022	2021	2020	2019	2022-2021	2021-2020	2020-2019
ASSETS							
Intangible assets	21220	12438	11809	11212	70,61%	5,33%	5,32%
Property, plant and equipment	31514	21885	21371	22749	44,00%	2,41%	-6,06%
Inventories	18679	14868	13391	14404	25,63%	11,03%	-7,03%
Trade receivables	3869	2076	1979	2228	86,37%	4,90%	-11,18%
EQUITY AND LIABILITIES							
Trade payables	12763	9650	7365	8814	32,26%	31,03%	-16,44%

BMW AUTOMOTIVE SEGMENT - Income Statement

INCOME STATEMENT (Automotive Segment)					HORIZONTAL ANALYSIS		
[MLN €]	2022	2021	2020	2019	2022-2021	2021-2020	2020-2019
Revenues	123602	95476	80853	91682	29,46%	18,09%	-11,81%
Profit / loss before financial result	10635	9870	2162	4499	7,75%	356,52%	-51,94%

Appendix 2

BMW GROUP - Indicators

INDICATOR	Formula	2022	2021	2020	2019
ROI	$\frac{EBIT}{TotalAssets - NonFinancialLiabilities}$	7.55%	7.50%	2.88%	4.20%
ROA	$\frac{EBIT}{Total Assets}$	5.67%	5.84%	2.23%	3.25%
ROS (EBIT Margin)	$\frac{EBIT}{Revenues}$	9.82%	12.05%	4.88%	7.11%
ATR	$\frac{Revenues}{Total Assets}$	57.75%	48.46%	45.69%	45.70%
ROE	$\frac{NetProfit}{Equity}$	20.36%	16.59%	6.27%	8.38%
RoCE	$\frac{EBIT}{(Equity + LongTermFinancialDebt)}$	9.67%	9.75%	3.75%	5.78%
DEBT TO EQUITY	$\frac{CurrentLiabilities + NonCurrentLiabilities}{Equity}$	1.705	2.055	2.522	2.806
QUICK RATIO	$\frac{CurrentAssets - Inventories}{CurrentLiabilities}$	0.855	0.919	0.930	0.905
CURRENT RATIO	$\frac{Current Assets}{Current Liabilities}$	1.092	1.127	1.137	1.097
CASH FLOW TO DEBT RATIO	$\frac{CashFlowsFromOperatingActivities}{CurrentFinancialLiabilities + NonCurrentFinancialLiabilities}$	24.97%	15.38%	12.46%	3.14%
CAPEX COVERAGE	$\frac{CashFlowsFromOperatingActivities}{CAPEX}$	2.599	2.404	2.155	0.531

BMW GROUP - Financial Leverage, DuPont Approach

INDICATOR	Formula	2022	2021	2020	2019
Net Profit Margin	$\frac{NetProfit}{Revenues}$	13.03%	11.20%	3.90%	4.82%
Asset Turnover Ratio	$\frac{Revenues}{Total Assets}$	57.75%	48.46%	45.69%	45.70%
Debt to Equity	$\frac{ThirdPartLiabilities}{Equity}$	1.705	2.055	2.522	2.806
ROE	$NPM \cdot ATR \cdot (1 + \frac{TPL}{Equity})$	20.36%	16.59%	6.27%	8.38%

BMW AUTOMOTIVE SEGMENT - Indicators

INDICATOR	Formula	2022	2021	2020	2019
RoCE	Values from the Annual Report	18.1%	24%	12.7%	29%
ROS	$\frac{EBIT}{Revenues}$	8.60%	10.34%	2.67%	4.91%

Appendix 3

AUDI GROUP AND MERCEDES GROUP - Indicators

INDICATOR	COMPANY	2022	2021	2020	2019
ROI	Audi	18.79%	15.99%	7.10%	12.25%
	Mercedes	9.92%	7.75%	3.05%	1.84%
ROA	Audi	10.66%	8.31%	3.82%	6.74%
	Mercedes	7.87%	6.17%	2.31%	1.43%
ROS	Audi	12.23%	10.36%	5.14%	8.10%
	Mercedes	13.64%	11.97%	4.28%	2.50%
ATR	Audi	87.21%	80.26%	74.33%	83.26%
	Mercedes	57.70%	51.53%	54.00%	57.10%
ROE	Audi	22.53%	21.72%	15.56%	13.89%
	Mercedes	17.11%	31.98%	6.44%	4.31%
DEBT TO EQUITY	Audi	1.242	1.542	1.772	1.355
	Mercedes	2.001	2.551	3.590	3.813
QUICK RATIO	Audi	1.200	1.177	1.093	1.107
	Mercedes	0.874	0.929	0.890	0.927
CURRENT RATIO	Audi	1.536	1.493	1.373	1.459
	Mercedes	1.164	1.174	1.155	1.208
CASH FLOW TO DEBT	Audi	1.165	1.372	0.529	0.890
	Mercedes	14.11%	18.37%	14.46%	4.60%
CAPEX	Audi	3.872	5.764	3.341	2.739
	Mercedes	2.449	3.354	2.609	0.728